1. **What do you think applying this filter to a grayscale image will do?** 1 point

$$\begin{bmatrix} -1 & -1 & 2 \\ -1 & 2 & 1 \\ 2 & 1 & 1 \end{bmatrix}$$

- (•) Detect 45-degree edges.
- ( ) Detect vertical edges.
- ( ) Detecting image contrast.
- ( ) Detect horizontal edges.

2. **Suppose your input is a 300 by 300 color (RGB) image, and you are not using a convolutional network. If the first hidden layer has 100 neurons, each one fully connected to the input, how many parameters does this hidden layer have (including the bias parameters)?** 1 point

- ( ) 9,000,100
- ( ) 27,000,001
- (•) 27,000,100
- ( ) 9,000,001

3. **Suppose your input is a 256 by 256 grayscale image, and you use a convolutional layer with 128 filters that are each 3×3. How many parameters does this hidden layer have (including the bias parameters)?** 1 point

- (•) 1280
- ( ) 75497600
- ( ) 1152
- ( ) 3584

4. **You have an input volume that is 63x63x16, and convolve it with 32 filters that are each 7x7, using a stride of 2 and no padding. What is the output volume?** 1 point

- ( ) 29x29x16
- (•) 29x29x32
- ( ) 16x16x32
- ( ) 16x16x16

5. **You have an input volume that is 61x61x32, and pad it using "pad=3". What is the dimension of the resulting volume (after padding)?** 1 point

- (•) 67x67x32
- ( ) 64x64x35
- ( ) 61x61x35
- ( ) 64x64x32

6. **You have a volume that is $121 \times 121 \times 32$, and convolve it with 32 filters of 5×5, and a stride of 1. You want to use a "same" convolution. What is the padding?** 1 point

- ( ) 3
- ( ) 0
- ( ) 5
- (•) 2

7. **You have an input volume that is 128x128x12, and apply max pooling with a stride of 4 and a filter size of 4. What is the output volume?** 1 point

- ( ) $64 \times 64 \times 12$
- ( ) $32 \times 32 \times 3$
- (•) $32 \times 32 \times 12$
- ( ) $128 \times 128 \times 3$

8. **Which of the following are hyperparameters of the pooling layers? (Choose all that apply)** 1 point

- [x] Filter size.
- [x] Whether it is max or average.
- [ ] Number of filters.
- [ ] Average weights.

9. **In lecture we talked about "parameter sharing" as a benefit of using convolutional networks. Which of the following statements about parameter sharing in ConvNets are true? (Check all that apply)** 1 point

- [x] It reduces the total number of parameters, thus reducing overfitting.
- [x] It allows parameters learned for one task to be shared even for a different task (transfer learning).
- [x] It allows a feature detector to be used in multiple locations throughout the whole input image/input volume.
- [ ] It allows gradient descent to set many of the parameters to zero, thus making the connections sparse.

10. **The sparsity of connections and weight sharing are mechanisms that allow us to use fewer parameters in a convolutional layer making it possible to train a network with smaller training sets. True/False?** 1 point

- (•) True
- ( ) False